

4.2.2 Plants and photosynthesis

The study of cells and transport into and out of cells in [Cells in animals and plants](#) (page 27) is developed and exemplified here in the context of plant science. A key part is the study of photosynthesis because this underpins work on the carbon cycle and climate change in [The Earth's atmosphere](#) (page 67) and the study of ecosystems in [Ecosystems and biodiversity](#) (page 75).

Plants can be attacked by bacteria and viruses so the successful growth of crops depends on methods to prevent or control infection.

There are two required practicals: one to investigate plant pigments by paper chromatography, another to investigate the effect of light intensity on the rate of photosynthesis.

4.2.2.1 Meristem tissue

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the function of meristems in plants.	Meristem tissue contains the cells in a plant that divide as the plant grows. This type of tissue is found at the growing tips of shoots and roots. The cells differentiate into different types of plant cells depending on where they are in the plant.	WS 1.4 Describe and explain the use of stem cells from meristems to produce clones of plants quickly and economically.

4.2.2.2 Plant structures

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe some of the substances transported into and out of a range of organisms, in terms of the requirements of those organisms, to include oxygen, carbon dioxide, water and mineral ions.	Plants, like other multicellular organisms, need specialised structures for transporting and exchanging materials. The roots, stem and leaves form a plant organ system for transport of substances around the plant. Plants take in carbon dioxide from the atmosphere for photosynthesis and oxygen for respiration. Plants take in water from the soil with dissolved ions including nitrate ions to make proteins and magnesium ions to make chlorophyll.	

4.2.2.3 Transpiration

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the need for exchange surfaces and a transport system in multicellular organisms. Explain how water and mineral ions are taken up by plants, relating the structure of the root hair cells to their function. Explain how the structure of xylem is adapted to its functions in the plant. Describe the process of transpiration including the structure and function of the stomata.	Water is drawn into the roots of plants from the soil. Water moves into the root hairs by osmosis. Mineral ions move from the soil into the root hairs by active transport. Water flows from the roots through xylem in its stems to its leaves. Xylem tissue is composed of hollow tubes strengthened by lignin adapted for the transport of water in the transpiration stream from the roots to the leaves. Water evaporates in the leaves and the water vapour escapes through tiny holes in the surface of leaves called stomata. The stomata can open or close as conditions change because the guard cells can gain or lose water by osmosis.	

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the effect of a variety of environmental factors on the rate of water uptake by a plant, to include light intensity, air movement and temperature.	The rate of transpiration varies with: • light intensity, which affects the opening of stomata • air movements, which affect the concentration of water vapour in the air around leaves • temperature, which affects the rate at which water evaporates.	MS 1a, 1c Understand and use simple compound measures such as the rate of a reaction. MS 4a Translate information between graphical and numerical form. MS 4a, 4c Plot and draw appropriate graphs, selecting appropriate scales for axes. WS 3.3 Carry out and represent mathematical and statistical analysis. MS 2c, 4a Extract and interpret information from graphs, charts and tables.

4.2.2.4 Chlorophyll and other plant pigments

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that chromatography involves a stationary and a mobile phase and that separation depends on the distribution between the phases. Interpret chromatograms, including measuring R_f values. Suggest chromatographic methods for distinguishing pure from impure substances.	The chlorophyll and other pigments in plant leaves can be separated and identified by chromatography. Chromatography can be used to separate mixtures and can give information to help identify substances. The ratio of the distance moved by a compound (centre of spot from origin) to the distance moved by the solvent can be expressed as its R_f value: $R_f = \frac{\text{distance moved by substance}}{\text{distance moved by solvent}}$ Different compounds have different R_f values in different solvents, which can be used to help identify the compounds. The compounds in a mixture may separate into different spots depending on the solvent, but a pure compound produces a single spot in all solvents.	MS 1a Recognise and use expressions in decimal form. MS 1c Use ratios and percentages. WS 3.3 Carry out and represent mathematical and statistical analysis. MS 1d Make estimates of the results of simple calculations. WS 4.6, MS 2a Use an appropriate number of significant figures. MS 4a Extract and interpret information from charts and tables. Translate information between graphical and numeric form when calculating R_f values.

Required practical activity 9: investigate how paper chromatography can be used to separate and tell the difference between coloured substances. Students should calculate R_f values.

AT skills covered by this practical activity: chemistry AT 1 and 4.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 167).

4.2.2.5 Photosynthesis

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the process of photosynthesis and describe photosynthesis as an endothermic reaction.	Photosynthesis takes place in the chloroplasts in the cells of the leaves of plants. The chloroplasts contain the chlorophyll, which absorbs sunlight. Photosynthesis is an endothermic reaction that can be represented by word and symbol equations. $\text{carbon dioxide} + \text{water} \xrightarrow{\text{light}} \text{glucose} + \text{oxygen}$ Energy is transferred to the plant cells by light. The glucose produced in photosynthesis may be: • used for respiration • converted into insoluble starch for storage • used to produce fat or oil for storage • used to produce cellulose, which strengthens the cell wall • used to produce amino acids for protein synthesis. To produce proteins, plants also use nitrate ions that are absorbed in solution from the soil.	WS 1.2 (HT only) Write a balanced symbol equation for photosynthesis given the formula of glucose.

4.2.2.6 Factors affecting the rate of photosynthesis

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the effect of temperature, light intensity and carbon dioxide concentration on the rate of photosynthesis.	The rate of photosynthesis depends on: • the energy available from light • the concentration of carbon dioxide in the air • the temperature.	MS 1a, 1c Carry out rate calculations for photosynthesis.

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
(HT only) Explain the interaction of these factors in limiting the rate of photosynthesis.	(HT only) The rate of photosynthesis may be limited by: • low temperature • shortage of carbon dioxide • shortage of light. (HT only) Increasing any one of the factors speeds up photosynthesis until the rate is limited by the factor which is in shortest supply.	WS 1.4 (HT only) Use data to relate limiting factors to the cost effectiveness of adding heat, light or carbon dioxide to greenhouses. MS 1a, 1c, 2c, 4a, 4c Translate information between numerical and graphical forms and extract and interpret information from graphs, charts and tables. WS 3.5 (HT only) Understand and use inverse proportion – the inverse square law – and light intensity in the context of factors affecting photosynthesis.

Required practical activity 10: investigate the effect of light intensity on the rate of photosynthesis using an aquatic organism such as pondweed.

AT skills covered by this practical activity: biology AT 1, 2, 3, 4 and 5.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 168).

4.2.2.7 Translocation

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the process of translocation. Explain how the structure of phloem is adapted to its functions in the plant.	Phloem tissue is composed of tubes of elongated living cells adapted for translocation of sugars from where they are produced by photosynthesis in the leaves to other parts of the plant for immediate use or storage. Cell sap containing sugars and other nutrients is able to move easily from one phloem cell to the next as the end walls have pores.	

4.2.2.8 Plant diseases

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how communicable diseases are spread in plants.	Tobacco mosaic virus is a widespread plant pathogen affecting many species of plants, including tomatoes. It gives a distinctive 'mosaic' pattern of discolouration on the leaves, which affects the growth of the plant due to lack of photosynthesis. Rose black spot is a fungal disease where purple or black spots develop on leaves, which often turn yellow and drop early. It affects the growth of the plant as photosynthesis is reduced. The disease is spread by spores of the fungus that are produced in the black spots.	
Explain how the spread of communicable diseases may be reduced or prevented in plants, to include a minimum of one plant disease.	Common control methods for tobacco mosaic virus include: • removing and destroying infected plants • washing hands and tools after handling infected plants • crop rotation to avoid planting in soil that has been infected for at least two years. Methods to control black spot include: • not planting roses too close together – to allow the air to flow freely around them • avoiding wetting the leaves when watering – wet leaves encourage the fungal disease • cleaning up any infected leaves from the ground round the roses – to avoid spores spreading • using a fungicide to prevent infection – spraying, especially in advance of warm, wet weather.	WS 1.4 Explain applications of science to prevent the spread of plant diseases.